

Physics-Constrained Bayesian Reconstruction of Unobserved Power-System States from Sparse PMUs

Working draft – Paper 2 of the sparse-PMU research program

Abstract—Sparse synchrophasor coverage turns transmission-system state estimation into a functional inverse problem: the individual operating-point perturbations that explain the measurements may not be unique, while the unobserved voltage field can remain nearly determined. We study this distinction on a validated PowerDynamics IEEE-39 benchmark with eight observed PMUs and 31 hidden buses. The estimator combines the nominal dynamic model with a constrained nonlinear AC maximum-a-posteriori (MAP) equilibrium solve. In the corrected M6 operating-point campaign, the nuisance vector has 43 coordinates and a median local rank of 25, yet the hidden-voltage static closure is 0.974, 0.981, and 0.983 for mismatch scales 0.5, 1.0, and 1.5, respectively. A subsequent causal streaming test shows that directly low-pass filtering dynamic PMU measurements does not provide a safe online center: online closure is negative, convergence is 0.8125, and nominal TVE increases materially. The results support functional, rather than parameter, identifiability as the central criterion for sparse-PMU reconstruction, while leaving causal center extraction and joint uncertainty calibration as open problems.

Index Terms—Power-system state estimation, sparse PMUs, functional identifiability, nonlinear AC MAP, operating-point mismatch, Bayesian estimation.

I. INTRODUCTION

Eight PMUs observe buses $\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}$ of the 39-bus system. The target is the hidden voltage field $V_{\mathcal{U}}$ on the 31 unobserved buses. This paper separates two questions that are often conflated: whether nuisance injections are uniquely identifiable and whether the hidden electrical field is identifiable as a functional of the data.

The paper makes four contributions. First, it formulates sparse-PMU reconstruction with explicit AC constraints and a physical nuisance basis. Second, it connects constrained local information diagnostics to hidden-voltage functionals. Third, it reports a reproducible mismatch campaign on a PowerDynamics-native IEEE-39 plant. Fourth, it documents the boundary between an effective static MAP recentering and an unsafe raw causal low-pass extension.

II. PROBLEM FORMULATION

Let $y_{\mathcal{P}} = h_{\mathcal{P}}(V) + v$ denote voltage and terminal-current phasors at the eight PMUs. The operating-point nuisance vector is

$$d = [\Delta P_{PQ}, \Delta Q_{PQ}, \Delta P_{PV}], \quad \dim d = 43,$$

with bus 31 as the slack and bus 39 as a PV bus. The estimator solves

$$\min_{V, d} \frac{1}{2} \|y_{\mathcal{P}} - h_{\mathcal{P}}(V)\|_{R^{-1}}^2 + \frac{1}{2} \|d - d_0\|_{\Gamma_d^{-1}}^2 \quad (1)$$

$$\text{s.t. } g_{AC}(V, d) = 0, \quad \theta_{31} = 0. \quad (2)$$

The hidden output is $V_{\mathcal{U}}$, not the nuisance vector itself.

III. FUNCTIONAL IDENTIFIABILITY

The constrained local map from nuisance coordinates to PMU outputs can have rank $r < 43$. E06-H gives median $r = 25$ and nullspace dimension 18. The relevant criterion is instead whether compatible nuisance perturbations are annihilated by the hidden-voltage functional, i.e., whether the constrained nullspace is approximately contained in the kernel of the hidden-output differential. This permits non-unique physical explanations with a stable reconstructed electrical field.

IV. VALIDATED BENCHMARK AND ESTIMATOR

The benchmark contains 114 differential and 78 algebraic states, uses the PowerDynamics IEEE-39 network primitives, and retains exact analytic AC and PMU Jacobians. Static parity, equilibrium, linearization, and measurement-operator audits precede the mismatch experiments. The physical nuisance basis keeps PV reactive power as a solved quantity and treats the slack explicitly.

V. EXPERIMENTS

A. Static M6 gate (E06-H)

Fresh DEV and TEST plants use the corrected basis, exact nominal network admittance, and a nonlinear AC MAP. The frozen TEST split contains 60 cases, 20 per nonzero mismatch scale. The nominal center, corrected MAP, oracle reference, AC residual, convergence, rank, and covariance are reported in the E06-H result files.

B. Causal streaming gate (E06-I)

Fresh streaming splits contain 40 DEV and 120 TEST trajectories, including a nominal $m = 0$ control. The DEV-only grid compares trailing means, a robust Huber trailing mean, and an exponential moving average over 10, 30, and 60 frames, with MAP updates every 1, 3, 10, or 30 frames. The selected causal configuration is evaluated without future samples or hidden truth.

TABLE I
CORRECTED STATIC M6 RECONSTRUCTION AND CAUSAL STREAMING
OUTCOME.

Mismatch	Nominal TVE	E06-H MAP TVE	E06-H closure
$m = 0.5$	0.19885%	0.00526%	0.974
$m = 1.0$	0.38834%	0.00749%	0.981
$m = 1.5$	0.57795%	0.00974%	0.983

VI. RESULTS

E06-H yields strong static recovery despite nuisance non-identifiability. The causal E06-I extension selected a 60-frame Huber window and 30-frame update cadence, but obtained negative online closure on all nonzero scales. The nominal control exposed the failure: R2 TVE rose to approximately 5.69%, while the frozen B2 reference remained below 0.02%. The failure is therefore consistent with dynamic-deviation leakage into the center target, not with a failure of the static AC MAP itself.

VII. UNCERTAINTY AND COMPUTATIONAL COST

The current Laplace approximation propagates static-center and Kalman deviation covariance while omitting their cross-covariance. Marginal coverage is not sufficient for calibration; the reported E06-H and E06-I NEES-like diagnostics remain poor. Static MAP solves take approximately 61.7 ms in E06-H; the selected E06-I update takes 38.3 ms median and 82.2 ms at p95. The latter is amortized to 2.31 ms per frame at a 30-frame cadence, but each update remains a blocking operation and is not a hard-real-time claim.

VIII. DISCUSSION

The evidence supports a hierarchy in which a nonlinear static AC MAP supplies the slow electrical center and a linear estimator supplies fast deviations. The E06-I result shows that the center cannot be obtained by low-pass filtering raw dynamic PMU channels. A causal innovation/deviation-compensated extractor is required before claiming an online adaptive estimator. Parameter identifiability is not the correct success criterion when the hidden voltage functional is stable across the nuisance nullspace.

IX. LIMITATIONS AND NEXT EXPERIMENT

The study uses one network and one PMU placement. Uncertainty is not yet calibrated, and the causal center problem remains open. The next experiment is a preregistered M6-only innovation-based center extractor, followed by a frozen transfer check to M1 and M7 only if the M6 nominal-safety gate passes.

X. CONCLUSION

Sparse PMUs can support physically constrained reconstruction of an unobserved voltage field under operating-point mismatch even when the nuisance injections are non-identifiable. The corrected static AC MAP recovers approximately 98% of the M6 mismatch error. The causal failure

is equally informative: raw low-pass centering is unsafe, so online adaptation must explicitly separate slow operating-point structure from fast dynamic deviations.

DATA AND CODE AVAILABILITY

Manifests, scripts, tests, and result tables are stored in the repository paths listed in the accompanying claim ledger. The generated physical trajectories remain evaluation artifacts; estimator inputs exclude hidden truth.

AUTHOR CONTRIBUTIONS, FUNDING, CONFLICTS, AND AI DISCLOSURE

To be completed before submission using the venue-specific declarations. No human or animal subjects are involved.